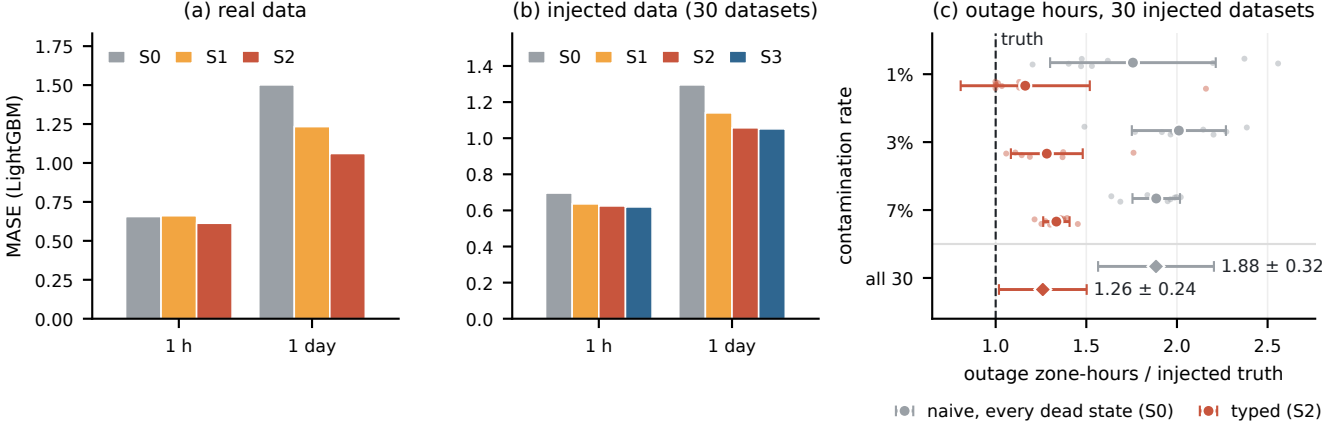
(a) real data
MASE (LightGBM)
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
S0
S1
S2
1 h
1 day
(b) injected data (30 datasets)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
S0
S1
S2
S3
1 h
1 day
(c) outage hours, 30 injected datasets
truth
contamination rate
1%
3%
7%
all 30
1.88 ± 0.32
1.26 ± 0.24
1.0
1.5
2.0
2.5
outage zone-hours / injected truth
naive, every dead state (S0)
typed (S2)